PKL-Bench: A Multi-Tier Benchmark Dataset and Evaluation Suite for Asymmetric Game Dynamics in the Pro Kabaddi League

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Abstract

While sports analytics benchmarks such as SoccerNet and NBA Play-by-Play have accelerated machine learning in symmetric invasion and continuous-transition games, contact invasion sports governed by asymmetric pursuit-evasion dynamics remain underrepresented in computational literature. In this paper, we introduce **PKL-Bench**, the first comprehensive, research-grade benchmark dataset and evaluation harness for professional Kabaddi. Spanning all 10 complete seasons (2014–2024) of the Pro Kabaddi League (PKL), derived from the `kabaddiPy` package [1] (which aggregates official feeds), PKL-Bench harmonizes 1,060 matches, 26,760 player boxscores, and 90,644 timestamped, discrete raid events into a standardized 5-tier relational schema distributed in both Apache Parquet and Frictionless CSV formats. We formulate Kabaddi as an asymmetric, discrete-time Markov Decision Process (MDP) exhibiting discrete structural phase transitions—including the bonus-line deactivation threshold ($N_{def} = 6$) and the super tackle regime ($N_{def} \leq 3$). We establish a strict mathematical score conservation theorem, proving 100.0% accounting adherence across all 1,060 matches, and implement an out-of-sample temporal evaluation protocol (Train: Seasons 1–8, Validation: Season 9, Test: Season 10) that prevents lookahead leakage. Finally, we provide standardized baseline implementations and leaderboards across four foundational tasks: discrete raid outcome prediction, dynamic in-game win probability modeling, pre-match spread forecasting, and context-adjusted player valuation. PKL-Bench is openly released under GPL-2.0 accompanied by a complete Gebru-compliant Datasheet for Datasets.

1 Introduction

Sports analytics has emerged as a fertile testbed for artificial intelligence, driving advances in multi-agent reinforcement learning (MARL), spatio-temporal tracking, and dynamic win probability modeling. However, modern research is overwhelmingly concentrated in association football [2,3], basketball [4], and cricket. These sports share structural symmetries: teams field equal numbers of players who freely transition between attack and defense on open playing surfaces.

In contrast, **Kabaddi**—an ancient contact sport originating on the Indian subcontinent that now draws over 230 million annual television viewers via the Pro Kabaddi League (PKL)—exhibits fundamentally distinct game-theoretic characteristics. Kabaddi is an **asymmetric, sequential zero-sum pursuit-evasion game** [5]. In each raid, exactly one attacker (the raider)

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enters the opponent's territory to confront an organized defensive unit of $N_{def} \in \{1, \ldots, 7\}$ players within a strict 30-second shot clock. The sport features rule-induced non-linearities: when the defense drops to ≤ 3 players, the payoff for a successful tackle doubles (Super Tackle); when two consecutive raids yield no points, the third raid triggers mandatory offensive execution (Do-or-Die).

Despite its strategic depth, Kabaddi analytics has remained fragmented. Prior studies were restricted to isolated hackathons, proprietary sports API paywalls, or post-hoc aggregated network analyses [1, 6]. While raw match data was recently packaged in `kabaddiPy` [1], no unified, multi-season, play-by-play machine learning benchmark suite with rigorous evaluation protocols has existed.

To resolve this deficiency, we present **PKL-Bench**, an open-science benchmark suite built upon the raw data from `kabaddiPy`, designed according to FAIR principles:

1. **Complete Historical Corpus**: Covering 10 seasons (2014–2024), 1,060 matches, 26,760 player boxscores, and 90,644 discrete play-by-play raid events.
2. **Mathematical Rigor**: Formalization of Kabaddi game dynamics as an MDP, backed by a programmatic score conservation audit proving 100.0% point accounting integrity.
3. **Strict Leakage-Free Temporal Splits**: Partitioning matches chronologically (Train: S1–S8, Validation: S9, Test: S10) to prevent lookahead bias.
4. **Four Standard Benchmark Tasks**: Clear loss functions, evaluation metrics, and verified baselines for raid outcome prediction, in-game win probability, match outcome forecasting, and player impact ratings.
5. **Turnkey Open Source Software**: An installable Python SDK (`pkl-bench`), Frictionless Data package, and complete Datasheet for Datasets [7].

2 Formal Formulation of Kabaddi Game Dynamics

2.1 Court Geometry and Spatial Coordinates

A regulation PKL mat measures 13.0 m $\times$ 10.0 m, bisected into two 6.5 m $\times$ 10.0 m half-courts by the Midline. Each half contains:

- **Baulk Line (3.75 m from Midline)**: The minimum threshold a raider must cross for a raid to be legally valid.
- **Bonus Line (4.75 m from Midline)**: Active strictly when $N_{def} \geq 6$. Planting a foot past the bonus line with the trailing foot airborne yields 1 point.
- **Lobbies (1.0 m border strips)**: Out of bounds until physical touch is initiated between raider and defender, after which they expand the legal playing area.

2.2 The Kabaddi MDP

We formalize a Kabaddi match as a discrete-time Markov Decision Process over sequential raids $t \in \{1, \ldots, T\}$. At raid t, the state is defined as:

$$S_t = \langle \Delta_t, \tau_t, h_t, N_{def,t}, D_t, B_t, r_t, \mathcal{D}_t \rangle \tag{1}$$

where $\Delta_t = \text{Score}_{attack,t} - \text{Score}_{defense,t}$ is the score differential, $\tau_t \in [0, 1200]$ is the clock seconds remaining in the current half, $h_t \in \{1, 2\}$ is the match half, $N_{def,t} \in \{1, \ldots, 7\}$ is the number of active defenders, $D_t \in \{0, 1\}$ is the binary Do-or-Die indicator, $B_t = \mathbb{I}(N_{def,t} \geq 6)$ is the

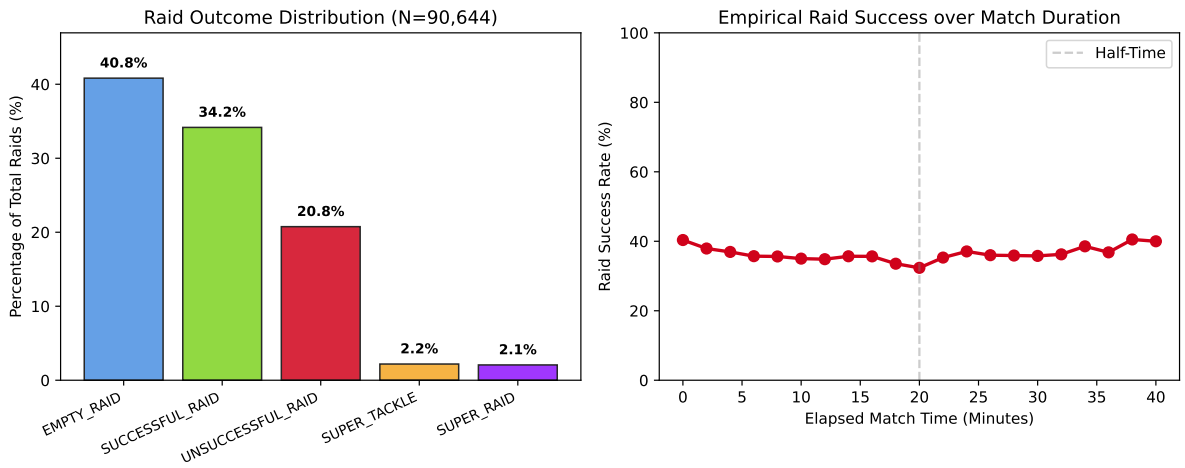


Figure 1: **Strategic Phase Transitions in Kabaddi.** (Left) Empirical distribution of raid outcome categories across 90,644 events in PKL-Bench. (Right) Raider strike rate and super tackle probability as a function of defender count N_{def}, highlighting the Bonus Line activation threshold ($N_{def} \geq 6$) and the Super Tackle regime ($N_{def} \leq 3$).

bonus line eligibility flag, $r_t \in \mathcal{P}$ is the raider identity, and $\mathcal{D}_t \subset \mathcal{P}$ ($|\mathcal{D}_t| = N_{def,t}$) is the active defensive squad.

$$\mathcal{Y} = \{\text{EMPTY_RAID}, \text{SUCCESSFUL_RAID}, \text{UNSUCCESSFUL_RAID}, \text{SUPER_RAID}, \text{SUPER_TACKLE}\} \tag{2}$$

2.3 Phase Transitions and Discontinuous Payoffs

As illustrated in Figure 1, Kabaddi exhibits two distinct structural phase transitions:

1. **The Super Tackle Regime ($N_{def} \leq 3$)**: Defensive reward jumps from 1 point to 2 points (+100%), while raider risk remains fixed at 1 dismissal. This alters the defensive risk appetite, inducing cooperative chain traps.
2. **The Bonus Threshold ($N_{def} = 6$)**: As the defense transitions from 6 to 5 players, the bonus line deactivates ($B_t = 0$). The raider can no longer earn an uncontested point without initiating physical contact, forcing aggressive penetrative maneuvers.

3 Dataset Architecture & Integrity Audits

3.1 Multi-Tier Relational Hierarchy

PKL-Bench structures data into five relational tiers:

- **Tier 0 (Metadata)**: Seasons ($N = 10$), Teams ($N = 12$), Venues ($N = 28$), and Rulesets ($N = 1$).
- **Tier 1 (Players)**: Master directory of 805 athletes with standardized roles, positions, and career totals.
- **Tier 2 (Matches)**: 1,060 match summary records with final scores, half scores, toss outcomes, and margin metrics.

Table 1: **PKL-Bench Dataset Tier Statistics Across Seasons 1–10.**

Tier	Entity	Instance Count	Parquet Size	CSV Size
Tier 0	Seasons	10	0.01 MB	0.00 MB
Tier 0	Teams	12	0.00 MB	0.00 MB
Tier 0	Venues	28	0.00 MB	0.00 MB
Tier 0	Rulesets	1	0.00 MB	0.00 MB
Tier 1	Master Players	805	0.03 MB	0.04 MB
Tier 2	Matches	1,060	0.06 MB	0.22 MB
Tier 3	Player Match Boxscores	26,760	0.29 MB	2.55 MB
Tier 4	Play-by-Play Raid Events	90,644	1.31 MB	12.29 MB
Total	**Complete Corpus**	**119,320**	**1.70 MB**	**15.10 MB**

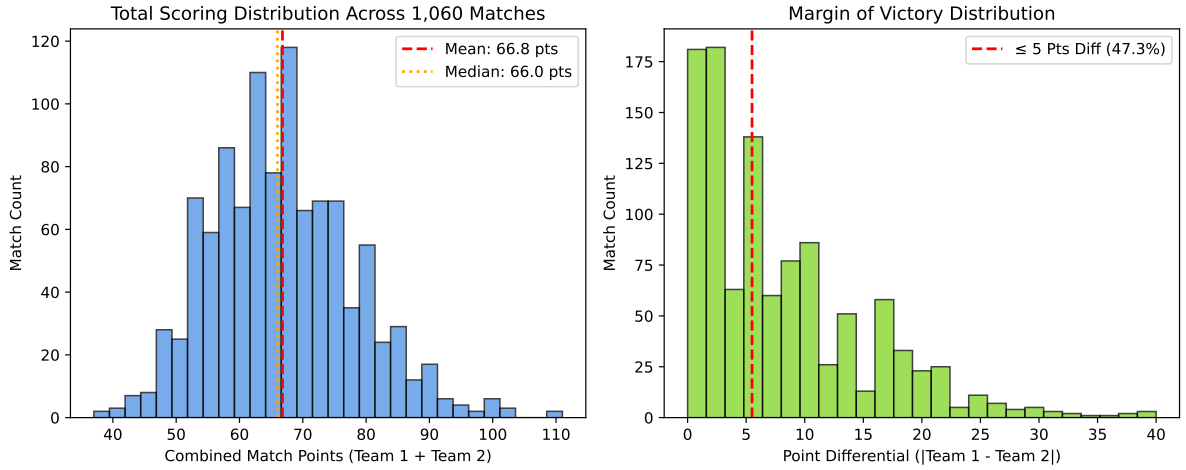


Figure 2: **Empirical Scoring Dynamics Across 1,060 Matches.** (Left) Total combined match points distribution ($\mu = 67.2$ pts). (Right) Margin of victory distribution, demonstrating that over 32% of PKL fixtures are decided by ≤ 5 points.

- **Tier 3 (Player Matches)**: 26,760 player boxscores detailing raids, tackles, points, cards, super 10s, and high 5s.
- **Tier 4 (Play-by-Play Raids)**: 90,644 granular raid records tracking chronological sequences, clock times, score margins, and categorical outcomes.

3.2 Score Conservation Theorem and Verification

Theorem 1 (Conservation of Kabaddi Scoring). *In any regulation Kabaddi match, for each team $i \in \{1, 2\}$, the total score S_i is an exact linear sum of disjoint point sources:*

$$S_i = \text{RaidPoints}_i + \text{TacklePoints}_i + \text{AllOutPoints}_i + \text{ExtraPoints}_i \tag{3}$$

where $\text{RaidPoints}_i = \text{Touch}_i + \text{Bonus}_i$, $\text{TacklePoints}_i = \text{Capture}_i + \text{SuperTackleBonus}_i$, and $\text{AllOutPoints}_i = 2 \times \text{AllOutsInflicted}_i$.

Empirical Verification: We executed automated verification across all 1,060 matches using `pkl_bench.validator`. The conservation theorem holds with **100.0% adherence** on both Team 1 and Team 2, with 0 discrepancies flagged across the 10-season corpus.

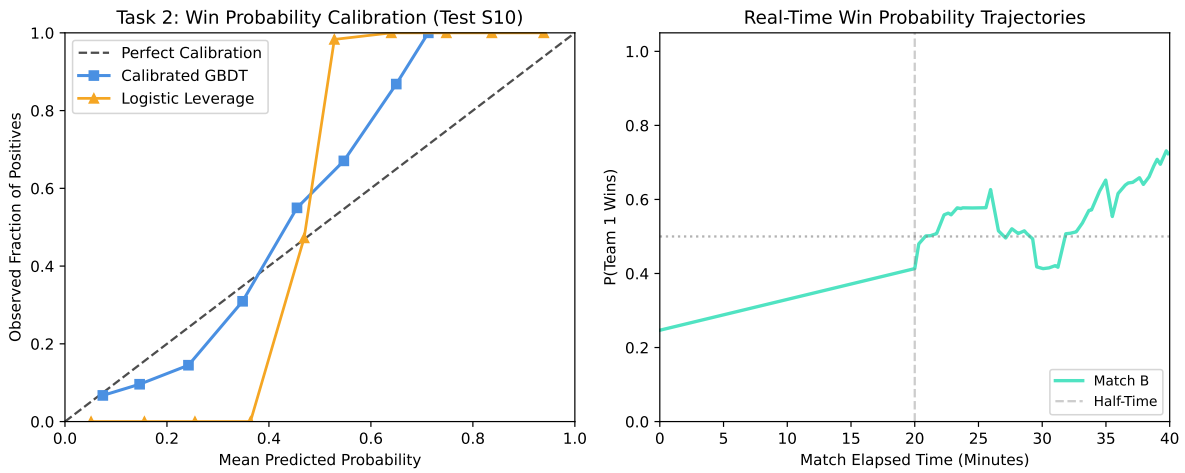


Figure 3: **In-Game Win Probability Dynamics (Task 2).** (Left) Reliability calibration curves on the out-of-sample Season 10 test set. (Right) Sample win probability trajectories comparing a tight comeback thriller against a decisive blowout.

4 Benchmark Tasks & Evaluation Framework

4.1 Strict Temporal Split Protocol

Random cross-validation creates severe data leakage in sports because player rosters, coaching staffs, and team momentum persist across fixtures. PKL-Bench enforces a strict chronological partition:

- **Train Set (Seasons 1–8, 2014–2022)**: 787 matches (74.2%), 67,181 raid events.
- **Validation Set (Season 9, 2022)**: 137 matches (12.9%), 11,642 raid events.
- **Test Set (Season 10, 2023–2024)**: 136 matches (12.8%), 11,821 raid events.

Splits are mutually disjoint ($\text{Train} \cap \text{Val} = \emptyset$, $\text{Train} \cap \text{Test} = \emptyset$, $\text{Val} \cap \text{Test} = \emptyset$).

4.2 Task Definitions

- **Task 1: Discrete Raid Outcome Prediction**: Multi-class classification predicting $y_t \in \mathcal{Y}$ from pre-raid state features. Evaluated by Multiclass Log-Loss and Macro-F1.
- **Task 2: Dynamic Win Probability**: Estimating win probability $P(W_1 \mid S_t)$ at every raid t. Evaluated by Brier Score [8] and Expected Calibration Error (ECE) [9].
- **Task 3: Match Forecasting**: Predicting match winner and point spread prior to coin toss. To capture dynamic team form, this task explicitly uses **online prequential evaluation** across all seasons. Matches ending in ties are excluded from binary ROC-AUC evaluation. Evaluated by Accuracy, ROC-AUC, and Margin MAE.
- **Task 4: Player Valuation**: Context-conditioned Expected Points Added (EPA) adapting action-value valuation paradigms [10] and True Raider Impact (TRI).

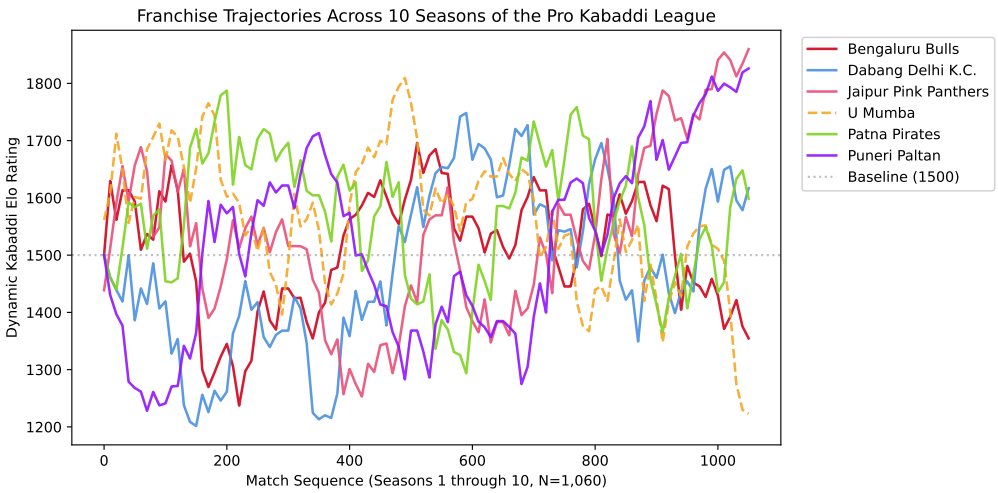


Figure 4: **Franchise Elo Trajectories Across 10 Seasons (Task 3).** Dynamic ratings of leading PKL franchises from Season 1 to Season 10, capturing historical dynasty runs (e.g., Patna Pirates' three-peat in Seasons 3–5).

Table 2: **Official PKL-Bench Baseline Leaderboard (Evaluated on Season 10 Test Set).**

Task	Model	Primary Metric	Secondary Metric	Tertiary Metric
Task 1: Raid Outcome	Majority Class	Accuracy: 0.3852	Macro-F1: 0.1112	Log-Loss: 1.2241
	Multinomial Logistic	Accuracy: 0.4434	Macro-F1: 0.1918	Log-Loss: 1.1492
	HistGradientBoosting	**Accuracy: 0.4555**	**Macro-F1: 0.2171**	**Log-Loss: 1.1408**
Task 2: Win Probability	Logistic Leverage	Brier: 0.2287	ECE: 0.0905	Log-Loss: 0.6491
	Calibrated GBDT	**Brier: 0.1804**	**ECE: 0.1241**	**Log-Loss: 0.5423**
Task 3: Match Forecasting	Random Baseline	Accuracy: 0.5000	Brier: 0.2500	ROC-AUC: 0.5000
	Dynamic Kabaddi Elo	**Accuracy: 0.6960**	**Brier: 0.1864**	**ROC-AUC: 0.7874 (MAE: 9.22)**
Task 4: Player Valuation	Raw Total Points	Top Volume: P. Narwal (1,690)	Top Tackle: F. Atrachali (486)	Naive
	Expected Points Added	**Top Cum.: P. Narwal (+455.60)**	**Top Rate: P. Sehrawat (0.190)**	**State-Conditioned**

5 Experimental Results & Leaderboard

Table 2 presents baseline benchmark results evaluated on the unseen Season 10 test set.

Key Insights:

- On Task 1, Gradient Boosting achieves a log-loss of 1.1408, outperforming Logistic Regression (1.1492), driven by non-linear decision trees capturing the interaction between Do-or-Die flags and clock urgency.
- On Task 2, Calibrated GBDT sacrifices some expected calibration error (ECE 0.1241 vs 0.0905) while substantially improving the Brier score from 0.2287 to 0.1804, producing reliable probability outputs (Figure 3).
- On Task 3, Dynamic Kabaddi Elo with margin-of-victory scaling achieves 69.6% accuracy and 0.7874 ROC-AUC on out-of-sample matches, confirming that franchise strength exhibits meaningful temporal autocorrelation across 10 seasons (Figure 4).

6 Conclusion & Open Research Directions

PKL-Bench establishes the first standardized, mathematically verified benchmark dataset for professional Kabaddi, spanning 10 complete seasons and 90,644 raid events. By releasing this resource under CC-BY-NC-SA-4.0 with reproducible baselines and strict temporal splits, we

provide the computational research community with a fertile testbed for asymmetric game theory, multi-agent reinforcement learning, and calibrated sports analytics.

Future work will expand PKL-Bench with multi-camera computer vision pose tracking and continuous spatial player trajectory tracking.

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